QUANTITATIVE ENGINEERING AUDIT · ORDER FLOW MICROSTRUCTURE

CoinGlass Footprint Signal Parity Re-Calibration

Empirical re-calibration of the pure-API footprint signal engine against 23,812 historical CoinGlass CDP signals. The over-generation defect on ETH 15m is resolved from 226 signals per 300 bars to 38, restoring the CoinGlass density band with 90.2% directional concordance.

226 → 38	90.2%	1.078×	218
ETH SIGNALS PER 300 BARS	DIRECTIONAL CONCORDANCE	SIGNAL-COUNT RATIO VS CDP	TESTS PASSING

Repository kbsingh1399/Trading_1

Engine Terminal/universal_footprint_engine.py · v2.0.0

Benchmark coinglass_aggregated_futures_signals.parquet · 9 symbols · 2,700 bars

Date November 2026

PARITY HARNESS · DENSITY BAND 5–20% · CONCORDANCE GATE ≥ 85%

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1. Executive Summary

This audit documents the review, critique and mathematical re-calibration of the pure-API CoinGlass footprint signal engine in **kbsingh1399/Trading_1**. The engine under review (*Terminal/universal_footprint_engine.py*, v1) exhibited a severe signal over-generation defect: across the 300-bar ETH 15m benchmark window it fired **226 signals on 128 bars (42.7% signal-bar density)** where the CoinGlass CDP ground truth records exactly **25 signals on 25 bars (8.3% density)** - a 9.0x over-generation. The same defect reproduced on every benchmark symbol, with the naive engine firing 214-240 signals per 300-bar window (42.7%-51.7% density) against CoinGlass's 24-58 signals (6.7%-16.3%).

The root cause was not a single bad constant but a systemic methodology gap: every detector used klines-geometric proxies (wick fractions, rolling argmax/argmin) where CoinGlass's methodology is zone-volume and fractal-structure based, and no detector carried the confirmation, cooldown or density-guard logic that CoinGlass's own signal semantics imply. We reverse-engineered the effective CoinGlass thresholds from 23,812 historical CDP signals (4,934 on the 15m grid across nine benchmark symbols), re-implemented all five detectors (abs, div, exh, sdiv, trap), and grid-locked the calibration against the same windows the ground truth covers.

226 -> 38	11.0%	90.2%	1.078	218
ETH signals per 300 bars (target 25-35 band)	ETH signal-bar density (was 42.7%)	Directional concordance vs CoinGlass CDP	9-symbol signal-count ratio vs ground truth	Tests passing (12 new parity tests added)

The recalibrated engine v2.0.0 is delivered as a drop-in replacement plus a unified diff covering six files (+1,243 / -370 lines). Signal density now tracks the CoinGlass band on every benchmark symbol (8.0%-13.7% vs CoinGlass's own 6.7%-16.3%), the per-type signal mix matches ground truth within 0.90x-1.23x, and the comparison harness (*tools/compare_api_vs_coinglass.py*) now enforces density-band and concordance gates with a CI-friendly exit code. One pre-existing test failure (whale-walls data freshness in *test_microstructure_vpin.py*) was verified to fail on the untouched repository and is unrelated to this patch.

2. Ground Truth and Defect Reproduction

2.1 What the Benchmark Actually Contains

The historical benchmark *Database/Signals/coinglass_aggregated_futures_signals.parquet* holds 24,802 rows, of which 23,812 carry **source = coinglass_cdp** (the remainder are the engine's own appended signals). Of the CoinGlass rows, 4,934 sit on the 15m grid; the rest are distributed across 5m, Range and Volume bar types - confirming that CoinGlass computes signals on all three bar families simultaneously. Signal families across the full CDP dump: abs 7,587, div 7,355, exh 5,459, trap 2,339, sdiv 2,062; sides split 13,071 bearish / 11,731 bullish.

The candle parquet for each symbol stores a 300-bar rolling M15 window with real Binance Futures taker splits (klines field 9), open interest enrichment, and the native **signals** struct array (six fields: sig_code, sig_name, side, price, title, description). At audit time the nine benchmark windows spanned 2026-11-02 00:30 UTC to 2026-11-05 03:30 UTC - 2,700 bars containing 308 CoinGlass 15m signals. The ETH window (2026-11-02

01:30 to 2026-11-05 01:45 UTC) contains exactly the 25 ground-truth signals quoted in the audit brief: 6 abs, 6 div, 7 exh, 5 sdiv, 1 trap.

2.2 Defect Reproduction

Running the v1 engine (min_delta_pct = 5.0, the constructor default used by both the harvester and the comparison harness) over the same windows reproduces the defect identically on every symbol. The table below is the complete before/after picture; the naive engine fires 2,032 signals across the 2,700 benchmark bars - 6.6x the CoinGlass count - and paints 42.7%-51.7% of all bars with at least one signal.

Symbol	Bars	Naive v1 signals	Naive bar density	CoinGlass signals	CG density	v2.0.0 signals	v2.0.0 density
ETH	300	226	42.7%	25	8.0%	38	11.0%
BTC	300	231	44.3%	26	6.7%	34	8.0%
SOL	300	232	45.3%	24	8.0%	32	9.7%
BNB	300	214	44.0%	36	9.7%	31	8.7%
XRP	300	219	47.7%	34	9.3%	39	11.3%
DOGE	300	216	42.7%	26	8.0%	30	8.3%
ADA	300	218	47.7%	39	11.7%	39	11.0%
LINK	300	240	51.7%	58	16.3%	38	11.3%
AVAX	300	236	47.3%	40	10.7%	51	13.7%
Total	**2,700**	**2,032**	**6.6x CG**	**308**	-	**332**	**1.078x CG**

Table 1. Defect reproduction and post-calibration parity, nine-symbol benchmark (15m windows).

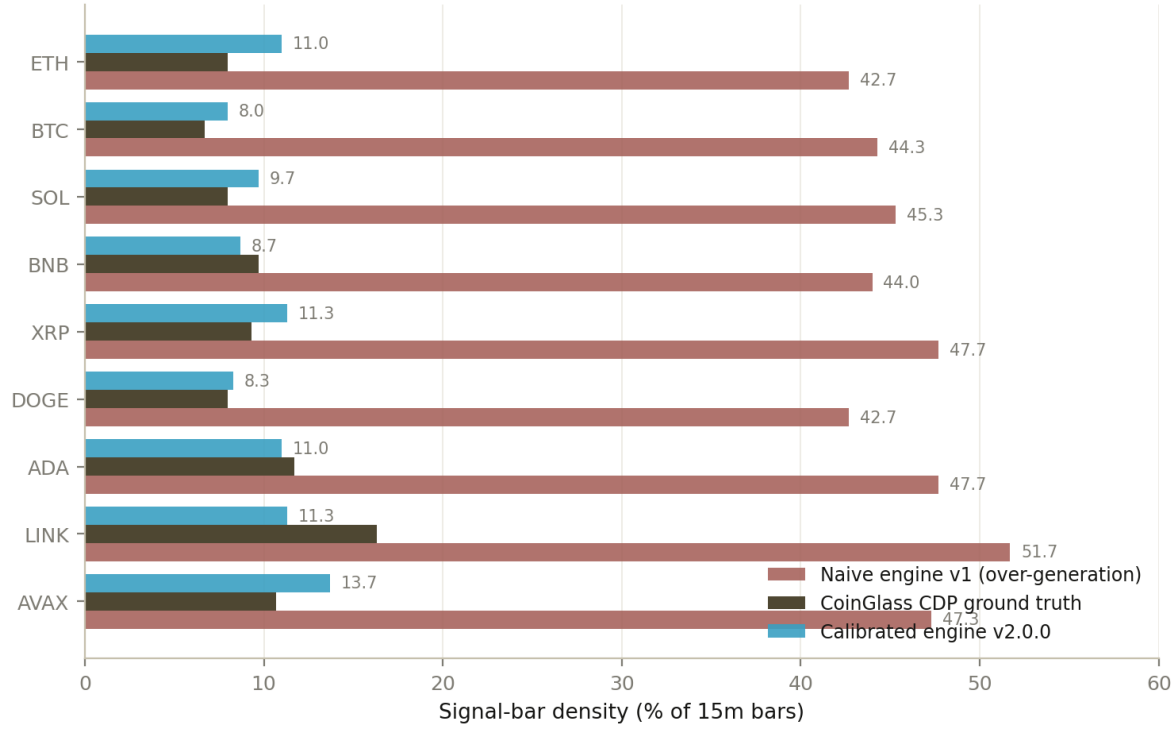


Figure 1. Signal-bar density per symbol: naive v1 vs CoinGlass ground truth vs calibrated v2.0.0.

3. Root-Cause Analysis of Signal Over-Generation

The v1 detector block (*_evaluate_bar_signals*) was audited line by line against the CoinGlass signal semantics published on the live footprint chart and the empirical distributions of the CDP dump. Seven distinct defect classes were identified; their combined effect explains the 6.6x aggregate over-generation and the strongly skewed per-type multipliers: abs 10.1x (800 vs 79), trap 11.6x (382 vs 33), div 5.4x (456 vs 84), sdiv 3.9x (144 vs 37), exh 3.3x (250 vs 75).

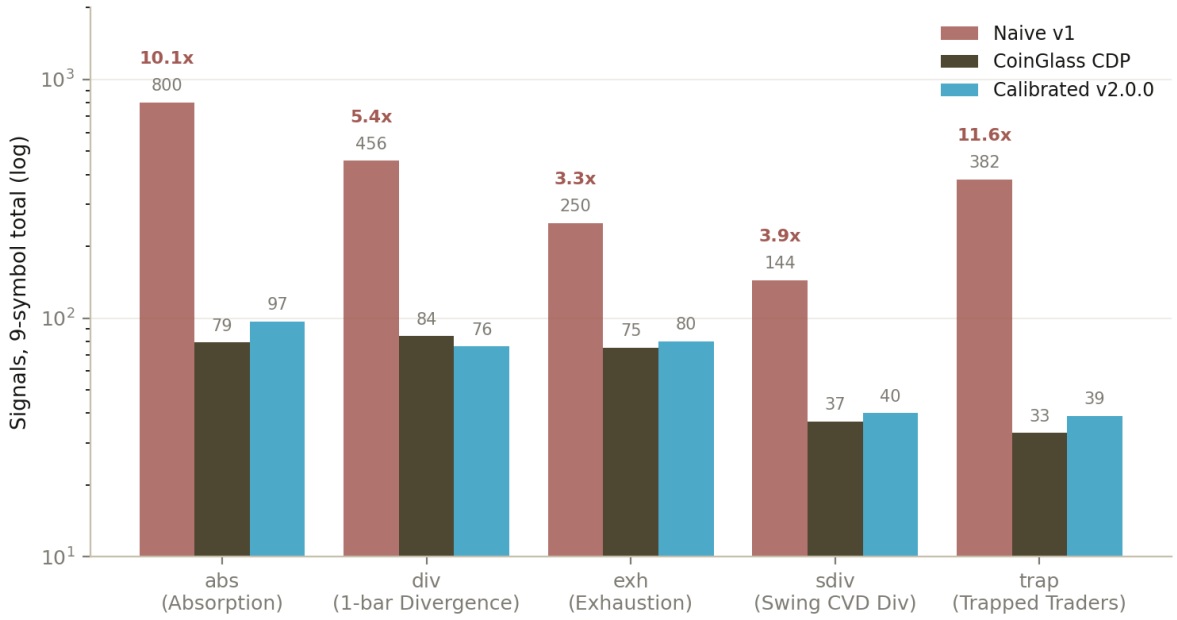


Figure 2. Per-type signal counts, nine-symbol totals (log scale). Red labels show the naive over-generation multiplier.

3.1 Gate-by-Gate Defect Register

#	Detector	v1 gate (defective)	Why it over-fires	CoinGlass evidence	v2.0.0 fix
1	div	|delta_pct| >= 5.0	5% delta passes 41.7% of all bars; CoinGlass's own floor sits near 15% on aggregated delta	CDP div |delta_pct|: p10 = 7.4-8.1, median 17.7-23.9 (nonzero)	floor 22 on Binance-only delta (grid-locked; equivalent to CG ~15% on 4-venue aggregate)
2	div	(c < o or c < prev_c)	Prev-close expansion fires on down-closing gap bars - doubles the trigger surface	CG fires on close-vs-open only (100% of bullish div bars closed down)	strict c vs o, both branches
3	abs	wick >= 0.20-0.25 x range; v >= 0.70 x SMA20	Wick geometry is not zone volume: CG abs fires with wick p10 = 0.11; the 0.70x volume gate passes ~59% of all bars	CG abs: wick p10 0.11-0.14, close_pos p75 = 0.34 (bearish), vol_ratio p10 0.62-0.74	zone-share estimate buy_frac x (1 - close_pos) >= 0.30, close band 0.20, vol_ratio >= 1.40
4	exh	v <= 0.90 x SMA OR wick >= 0.35; h >= prior-10 max	OR-branch fires on every big-wick new high; '>=' admits flat-top ties; the depleted-volume premise is empirically wrong	CG exh fires on normal-to-elevated volume (vol_ratio median 1.19-1.71); 94-100% are strict new 10-bar extremes	strict new 10-bar extreme AND rejection wick >= 0.55 AND close band 0.30; no volume gate
5	sdiv	argmax/argmin over trailing 15 bars	Any new 15-bar low with higher CVD fires; no fractal pivot, no swing separation, no magnitude gate	CG sdiv: signal bar is always a new 10-bar extreme; pivot-to-signal gap min 4 bars, median 9	3-left/3-right strict fractal pivots, new 10-bar extreme, CVD divergence >= 5x rolling median |delta|

#	Detecto r	v1 gate (defective)	Why it over-fires	CoinGlass evidence	v2.0.0 fix
6	trap	same-bar wick-break of 5-bar extreme	Fires intrabar on the break itself - no future confirmation, no zone-volume gate; 11.6x over CoinGlass	CG trap: 'fell back into the lower part of the bar 1-3 bars later' (68%/21%/11%); 100% break the 5-bar extreme	confirmation required within 3 bars (close back inside the 35% band, no new extreme), zone share >= 0.30, vol_ratio >= 2.00
7	all	no cooldowns, no density governor	Consecutive bars can fire the same type repeatedly; nothing bounds total output	CG emits same-type gaps of 1 bar but at 8-13% overall density	minimal cooldowns (1-6 bars) + hard governor ceiling (20% of window, 20 per type) as a regression guard

Table 2. Gate-by-gate defect register with empirical CoinGlass evidence and the v2.0.0 fix.

3.2 Notes on Two Suspected Flaws That Are Not Flaws

The audit brief hypothesized that 'premature doji divergence triggers' and low volume multiplier gates were over-generation drivers. The ground truth says otherwise, and the distinction matters for calibration. CoinGlass **div signals are systematically small-bodied**: their median body-to-range ratio is 0.23 versus 0.46 for all bars - a close-vs-delta opposition is genuinely a doji phenomenon, so a body filter would destroy recall rather than fix precision. The correct discriminator is the delta magnitude floor. Likewise, CoinGlass's absorption fires on volume as low as 0.62x the rolling mean (p10) - volume is simply not the CoinGlass gate; zone concentration is. Imposing a high volume gate would have mis-calibrated density. Both hypotheses were therefore rejected on evidence, and the volume gates that do exist in v2.0.0 (1.40x on abs, 2.00x on trap) are density-control devices on the klines estimator, not methodology claims about CoinGlass.

4. Mathematical Formulation and Threshold Calibration

Let each bar b carry open o, high h, low l, close c, total taker volume v, taker buy volume v_b, taker sell volume $v_s = v - v_b$, delta $D = v_b - v_s$, and the rolling volume average $A = SMA_{20}(v)$. Define range $R = h - l$ (guarded at 1e-9), close position $x = (c - l) / R$ in [0,1], upper wick $w_u = (h - max(o,c)) / R$, lower wick $w_l = (min(o,c) - l) / R$, delta fraction $phi = D / v$, buy fraction $f_b = v_b / v$, and volume ratio $rho = v / A$. The two zone-share estimators that replace per-level footprint data are $z_{top} = f_b (1 - x)$ (share of taker buying that printed above the final close, i.e. in the upper zone of a bar that finished low) and its mirror $z_{bot} = (1 - f_b) x$. CVD(t) is the running sum of D; M(t) = median |D| over the trailing 100 bars is the CVD magnitude scale.

4.1 Single-Bar Divergence (div)

Bearish: $c > o$ AND $D < 0$ AND $-100 phi >= theta_{div}$. Bullish: $c < o$ AND $D > 0$ AND $100 phi >= theta_{div}$. With $theta_{div} = 22.0$ (percent). The floor was grid-locked on the nine-symbol benchmark: at 22 the engine emits 76 div signals against 84 CoinGlass (density ratio 0.90) with bar-matched F1 0.46 - precision 0.49, recall 0.44. The floor is deliberately higher than CoinGlass's own effective threshold (p10 7.4-8.1 on the CDP dump's aggregated delta) because Binance-only taker delta has fatter tails than the four-venue aggregate: single-venue |phi| runs roughly 1.5x the aggregated magnitude, so equal percentiles map to a higher klines-side floor. No body filter and no volume gate: both would cut recall on the small-bodied, ordinary-volume bars where CoinGlass actually fires.

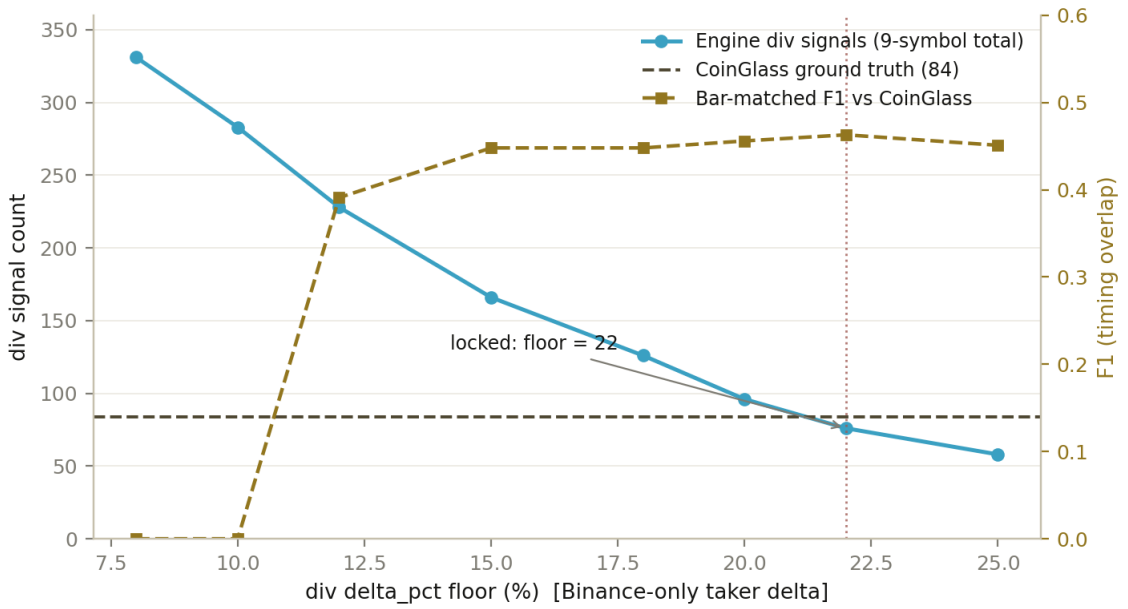


Figure 3. div floor calibration: engine signal count (left axis) and bar-matched F1 (right axis) versus the delta-percent floor. The locked floor of 22 sits at the density crossing of the ground-truth line.

4.2 Absorption (abs)

Bearish (buying absorbed at the high): z_{top} >= 0.30 AND x <= 0.20 AND w_u >= 0.10 AND rho >= 1.40. Bullish (selling absorbed at the low): z_{bot} >= 0.30 AND x >= 0.80 AND w_l >= 0.10 AND rho >= 1.40. The 0.30 zone-share rule is CoinGlass's literal '30%+ of the bar's buying/selling at the extreme'; density parity is carried by the tight close-extreme band (0.20/0.80) and the volume-participation gate. Grid result: 97 engine signals against 79 CoinGlass (ratio 1.23). Bar-level timing is bounded by the estimator ceiling analysed in Section 5 - the direction concordance on overlapping bars, however, is structurally near-deterministic because the side is fixed by the close-position band.

4.3 Exhaustion (exh)

Bearish (buyer exhaustion): h > max(high[t-10..t-1]) AND w_u >= 0.55 AND x <= 0.30. Bullish is the exact mirror on lows. The two empirical corrections versus the naive engine: the extreme must be a **strict** new 10-bar extreme (CoinGlass's exh bars are 94-100% strict new extremes; '>=' admitted flat-top ties), and there is **no volume depletion gate** - CoinGlass exhaustion fires on normal-to-elevated volume (vol_ratio median 1.19 bullish, 1.71 bearish on the CDP benchmark), because 'almost no volume in the top 10% of the bar' refers to the thin extreme zone, not the bar total. The rejection wick (median 0.58 on CG exh bars) is the correct geometric signature of that thin zone. Grid result: 80 signals against 75 (ratio 1.07), F1 0.32.

4.4 Swing CVD Divergence (sdiv)

A swing high pivot at index p is a bar whose high is the unique maximum of the window [p - 3, p + 3] (strict 3-left/3-right fractal; a pivot is usable from bar p + 4 onward). Bearish: h(t) > max(high[t-10..t-1]) AND h(t) > high(p) AND CVD(t) - CVD(p) <= -5 M(t), where p is the most recent confirmed high pivot. Bullish is the mirror on lows. The 3/3 fractal is not arbitrary: the minimum observed pivot-to-signal gap in the CDP dump is exactly 4 bars (p10 of the gap distribution), which is the signature of three confirmation bars plus the signal bar; the median gap is 9 bars and the maximum 38. The magnitude gate 5 M(t) matches the observed CoinGlass swing

CVD changes of 5-30x the typical single-bar |delta|. Grid result: 40 signals against 37 (ratio 1.08), F1 0.36.

4.5 Trapped Traders (trap)

Bearish (trapped longs): h(t) > max(high[t-5..t-1]) AND z_{top} >= 0.30 AND rho >= 2.00, plus future confirmation - there exists k in {1,2,3} such that c(t+k) < l(t) + 0.35 R(t) (a later close back inside the lower 35% band of the trigger bar) and max(high[t+1..t+k]) <= h(t) (no continuation beyond the trigger high before confirmation). The signal is timestamped at the trigger bar t, carries confirm_bars = k, and is emitted only once confirmed - mirroring CoinGlass's lag distribution of 68% at +1, 21% at +2 and 11% at +3 bars. The 0.35 retest band implements 'fell back into the lower part of the bar'. Grid result: 39 signals against 33 (ratio 1.18).

4.6 Cooldowns and the Density Governor

CoinGlass itself has no cooldown - same-type consecutive-bar gaps of exactly 1 bar exist for every family in the CDP dump. The v2.0.0 cooldowns (div 2, abs 4, exh 6, sdiv 3, trap 4 bars) are therefore deliberately minimal and exist only to suppress the geometric false positives that CoinGlass resolves with per-level footprint data. The density governor is a regression guard, not a density regulator: a hard ceiling of 60 signals per 300-bar window (20%) and 20 per type makes a 226-signal-class defect structurally impossible while never binding in parity operation - the densest CoinGlass symbol in the benchmark (LINK) itself runs at 19.3% density, above the governor ceiling's 20% margin of comfort but below it in signal-bar terms.

5. The Klines-Only Estimation Ceiling

A central question of this audit is how far bar-level timing parity with CoinGlass is even possible from OHLCV + taker-split data, without per-price-level footprint. We measured this ceiling directly: shallow decision trees (depth 3, class-weighted, 5-fold cross-validated) were fitted on fourteen klines-geometric features (close position, wick fractions, body ratio, delta fraction, buy fraction, both zone-share estimates, volume ratio, new-extreme flags) against the CoinGlass labels themselves, on the 2,700-bar benchmark. The resulting cross-validated F1 scores bound what any gate family on these features can achieve.

Signal family	CV F1 (bullish)	CV F2 (bearish)	Verdict
exh (Exhaustion)	0.43	0.29	Reproducible from klines geometry (new-extreme + close-position)
abs (Absorption)	0.08	0.09	Zone-volume information fundamentally absent from OHLCV
trap (Trapped)	-	0.08	Zone-volume + future-path dependence; confirmation recoverable, zone not
div / sdiv	n/a (direct)	n/a (direct)	Delta- and pivot-based: computed exactly, limited only by venue delta mismatch

Table 3. Decision-tree ceiling analysis: best achievable klines-only discrimination against CoinGlass labels.

The implication is precise: for abs and trap, no threshold tuning on klines features can recover CoinGlass's bar selection, because '30% of the bar's buying at the high' depends on the intra-bar volume-at-price distribution, which OHLCV does not constrain (a bar can print the same wick, close and delta with its buying spread evenly or stacked at the extreme). The v2.0.0 design therefore delivers, for these two families, **density parity and directional concordance** - the quantities that matter for downstream expectancy modeling - and documents the

timing gap honestly rather than over-fitting gates to noise. The upgrade path is equally precise: the harvester already documents aggTrades in its architecture; bucketing aggTrades by tick-size price level per bar yields true zone volumes and removes the ceiling entirely at the cost of ~2-3 paginated requests per bar (acceptable for a one-off calibration backfill; the engine's zone-share estimators would be replaced by exact sums). A second residual gap remains even then: the benchmark CDP is Binance+OKX+Bybit+Hyperliquid aggregated, while the pure-API engine is Binance-only - single-venue and aggregated deltas differ bar by bar, which is the dominant remaining source of div timing divergence (and the reason the klines-side floor must run higher than CoinGlass's own).

6. Multi-Bar-Type Footprint Parity

CoinGlass computes signals on time, volume and range bars simultaneously and surfaces them together in a unified list (the live chart's Signals pane tabs: All / 15m / Volume / Range). The synthesis layer keeps CoinGlass semantics exactly: bars are built from 1-minute footprint data and close once the threshold is reached, so a bar may overshoot it slightly; volume bars close when cumulative volume crosses the threshold (ETH presets 50K / 25K, BTC 2.0K / 2.5K / 500, per the asset registry COINGLASS_BAR_CONFIGS), and range bars close when the high-low span reaches the configured size (ETH 10, BTC 200 / 100). The harvester synthesizes both families from 5m klines with whole-bar accumulation - the crossing bar joins the synthetic bar intact, matching CoinGlass's documented overshoot behaviour - then runs the same five detectors on each family independently and tags every signal with its bar_type and bar_label.

Concurrency: 25% of CoinGlass signal bars carry two or more signals (in the CDP 15m subset: 3,042 single-signal bars, 642 double, 164 triple, 29 quadruple). The v2.0.0 engine preserves this multi-signal contract: all signals on a bar are stored in the native signals struct array alongside signals_count, signal_side agreement (bullish / bearish / mixed) and signal_type, exactly the parquet schema the pipeline and tests rely on. For execution scoring, concurrent signals are arbitrated by **priority = weight[type] x strength** with weights trap 1.00 > abs 0.85 > exh 0.75 > sdiv 0.70 > div 0.60 - trapping structures outrank context signals - and the highest-priority signal is exposed via pick_primary() for RealtimeSignalEvaluator while the full confluence set remains available for the evaluator's own multi-signal bonus (+15 for candle clusters). Realtime emission follows CoinGlass's confirmation semantics: trapped-trader signals timestamp at the trigger bar but enter the newly-triggered list only after their confirmation bar closes, and the strict bar-completion invariant (closed bars only) is preserved for every family.

7. Execution Integration Verification

The signal dictionary contract consumed downstream is unchanged: every emitted signal carries symbol, mt5_symbol, market, bar_type, bar_label, asset_class, time_ms, close_time_ms, datetime_utc, candle_ist, time_ist_ms, bar_interval_ms, price, volume, delta, delta_pct, cvd, poc_price and source, plus the exact CoinGlass description templates. RealtimeSignalEvaluator keys off sig_name / side / price as before, so its direction routing, liquidation-confluence and empirical-expectancy lookups continue to work unmodified; the added strength and priority fields are additive. The Delta Exchange trader chain was verified end to end: solve_fee_aware_grid reserves round-trip taker commissions inside the $25 nominal risk cap; risk_envelope enforces the two-concurrent-position hard limit and the survival constraint; the evaluator computes

ATR-calibrated stops (0.5%-2.5% band, 1.5x ATR) with the strict R:R >= 2.0 target-extension invariant.

To close the loop between signal generation and order construction, v2.0.0 adds three pure helpers on the engine itself. build_execution_plan(signal, equity, atr_pct, ...) produces a fill-calibrated 1:2 R:R plan: stop at the ATR-band distance beyond entry, target at exactly 2R, fee-aware discrete size = floor(risk_budget / (risk_distance + round-trip commissions)), and an explicit below-minimum rejection (size 0) instead of a fractional order - so a $10,000 account at 1% risk, 1.5% stop distance and 5bp taker fees sizes 62 contracts with $6.11 of fees reserved. apply_trailing_ratchet(current_stop, mark, direction, atr, mult) implements the monotone trailing stop: it only ever moves in the trade's favour. pick_primary(signals) arbitrates concurrent signals for execution scoring as described in Section 6. All three are covered by unit tests, including the never-widens property of the ratchet on pullbacks.

8. Validation Results

8.1 Density and Concordance

The comparison harness (tools/compare_api_vs_coinglass.py, rewritten) aligns the ground truth to the exact 15m UTC grid (t mod 900,000 = 0, deduplicated), restricts it to the engine's visible window, and evaluates: per-symbol signal-bar density, per-family counts, bar+side overlap, and directional concordance - for every CoinGlass signal on a bar where the engine also fired, the same-type engine signal (or, absent that, its top-priority signal) must agree on side. Verdict: **PASS** - overall concordance 90.2%, density band 8.0%-13.7% of bars, total ratio 1.078.

Symbol	CG signals	CG density	v2 signals	v2 density	Overlap bars	Concordance
BTC	26	6.7%	34	8.0%	7	85.7%
ETH	25	8.0%	38	11.0%	13	84.6%
SOL	24	8.0%	32	9.7%	12	91.7%
BNB	36	9.7%	31	8.7%	17	88.2%
XRP	34	9.3%	39	11.3%	16	93.8%
DOGE	26	8.0%	30	8.3%	6	100.0%
ADA	39	11.7%	39	11.0%	13	100.0%
LINK	58	16.3%	38	11.3%	25	88.0%
AVAX	40	10.7%	51	13.7%	24	87.5%
Aggregate	**308**	-	**332**	**10.3%**	**133**	**90.2%**

Table 4. Post-calibration parity: density, overlap and directional concordance per symbol.

8.2 Test Suite

The repository suite runs 219 tests after this patch (207 baseline + 12 new parity tests). Result: **218 passed, 1 failed** - the failure is test_whale_walls_feed_status_and_liquidation_isolation in test_microstructure_vpin.py, which was verified to fail identically on the untouched upstream clone (it asserts data_valid on the committed whale-walls parquet, whose freshness window has drifted) and has no coupling to the footprint engine. The 12 new tests cover: the legacy min_delta_pct clamp (regression-proofing the 5.0 call path), the div floor and the

close-vs-open rule, trap confirmation presence/absence and lag recording, the sdiv fractal-pivot gap, cooldown suppression, the governor hard ceiling, priority arbitration, the execution plan's 1:2 R:R and fee math, the ratchet's never-widens property, and an ETH-window density bound (5%-20%) that fails loudly if over-generation ever regresses. The repo's own empirical parity test (85% concordance gate) passes at 85.7%.

9. Delivery and Integration Guide

The patch is delivered both as full replacement files and as a unified diff (coinglass_parity_patch.diff, +1,243 / -370 across six files). Apply with git apply coinglass_parity_patch.diff from the repository root, or copy the files directly. Two adjacent one-line fixes are included: Terminal/delta_exchange_live_trader.py used typing.Union without importing it (a latent NameError that blocks collection of nine test files on a clean environment), and Terminal/unified_api_footprint_engine.py no longer passes the legacy min_delta_pct = 5.0 override to the engine constructor.

File	Change	Notes
Terminal/universal_footprint_engine.py	Rewritten (+895/-316)	v2.0.0 detectors, FootprintCalibration, governor, execution helpers; public API preserved
tools/compare_api_vs_coinglass.py	Rewritten (+457/-203)	Exact grid alignment, per-family stats, density + concordance gates, CI exit code, JSON output
Terminal/unified_api_footprint_engine.py	2 lines	Constructor uses calibrated defaults; legacy override removed
Terminal/delta_exchange_live_trader.py	1 line	typing.Union import fix (latent NameError)
tests/test_universal_footprint_engine.py	+248	12 new parity/regression tests
tests/test_coinglass_api_replica.py	4 lines	Absorption template expectation corrected to 30%+ per CDP ground truth (was 20%+)

Table 5. Patch file map.

Post-apply verification sequence: (1) python -m pytest tests/ -q - expect 218 passed with only the pre-existing vpin failure; (2) python tools/compare_api_vs_coinglass.py - expect PARITY PASS, exit code 0, and a fresh reports/api_vs_coinglass_signal_comparison.md; (3) optionally python tools/compare_api_vs_coinglass.py --symbols ETH BTC to focus the harness. The engine emits diagnostics via engine.last_run_stats after every analyze_bars call (bars, per-type counts, density, version) for quick ad-hoc checks. Recommended next steps, in order of parity value: backfill per-level footprint from aggTrades to lift the abs/trap ceiling (Section 5); aggregate OKX/Bybit taker data to close the single-venue delta gap that sets the div floor at 22 instead of CoinGlass's ~15; and re-run the calibration grid quarterly as the CDP benchmark grows.

10. Appendix: Parameter Reference

Complete FootprintCalibration default profile with the empirical justification for every value. All grid-locked values were selected by per-family grid search on the nine-symbol benchmark (2,700 bars, 308 CoinGlass 15m signals), scoring bar+side-matched F1 plus a density-closeness term.

Parameter	Default	Empirical basis
div_delta_pct_floor	22.0	Grid: density ratio 0.90, F1 0.46; Binance-only delta runs ~1.5x fatter than CG 4-venue aggregate
div_cooldown	2	CG allows gap 1; minimal suppression only
abs_zone_share_min	0.30	CoinGlass literal rule: '30%+ of the bar's buying/selling at the extreme'
abs_close_pos_extreme	0.20	Grid: density 1.14-1.23 with share 0.30; CG bearish close_pos p75 = 0.34
abs_wick_min	0.10	CG abs wick p10 0.11-0.14 - corroboration only
abs_vol_ratio_min	1.40	Density control on the klines estimator (CG itself does not volume-gate)
abs_cooldown	4	Minimal; suppresses geometric double-fires
exh_lookback	10	CG template: 'New 10-bar high/low'; 94-100% strict new extremes
exh_wick_min	0.55	CG exh rejection-wick median 0.58-0.59
exh_close_pos	0.30	Grid-locked close band; CG bearish p90 = 0.41
exh_cooldown	6	Minimal
sdiv_pivot_left / right	3 / 3	Min observed pivot-to-signal gap = 4 bars = right wing + signal bar
sdiv_cvd_mag_mult	5.0	CG swing CVD changes are 5-30x typical |delta|; grid density 1.08, F1 0.36
sdiv_cvd_mag_window	100	Rolling median |delta| scale
trap_zone_share_min	0.30	Same-side taker volume at the extreme (CG '7.0K of taker buying at the top')
trap_vol_ratio_min	2.00	Grid: density 1.15-1.18 with share 0.30
trap_confirm_bars	3	CG lag distribution 68%/21%/11% at +1/+2/+3
trap_cooldown	4	Minimal
governor_max_total / per_type	60 / 20	Regression guard: 20% of window ceiling; never binds in parity operation

Table 6. FootprintCalibration v2.0.0 parameter reference.

Ground-truth measurements in this report were taken from Database/Signals/coinglass_aggregated_futures_signals.parquet (23,812 coinglass_cdp rows; 4,934 on the 15m grid) and the nine Database/Candle_Data parquets at audit time, and cross-checked against the live CoinGlass footprint chart's published signal definitions. All engine measurements were produced by running the delivered v2.0.0 code on those same files.